Why impact also needs energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why momentum does not tell the whole impact story.

SUCCESS CRITERIA - I can...

1. Compare momentum and kinetic energy as different quantities.
2. Identify speed as a squared factor in kinetic energy.
3. Name where energy can go in an impact.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A and B have equal momentum magnitude, but A has 3 times B's mass. Compare speeds and kinetic energies algebraically.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p = mv$; $K = \frac{1}{2}mv^2$.
Momentum includes direction; kinetic energy is a scalar.
Neither alone gives a complete damage forecast.

Worked reasoning: A 2 kg cart at 3 m/s has momentum 6 kg·m/s and kinetic energy $\frac{1}{2}\times2\times3^2=9$ J. A 1 kg cart at 6 m/s has the same momentum but 18 J of kinetic energy.

Your turn: Draw labeled momentum arrows and energy bars for the two carts.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Two equal-mass carts move at speeds 2 and 4 m/s. Predict how their momenta and kinetic energies compare before calculating.



Same mass; 2 m/s Same mass; 4 m/s

Marzano check - circle: ① ② ③

04 Apply the physics

Two lab carts each have momentum magnitude 12 kg·m/s. Their masses are 2 kg and 6 kg. Find each speed and kinetic energy. Draw energy bars and explain why equal momentum does not guarantee equal kinetic energy.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

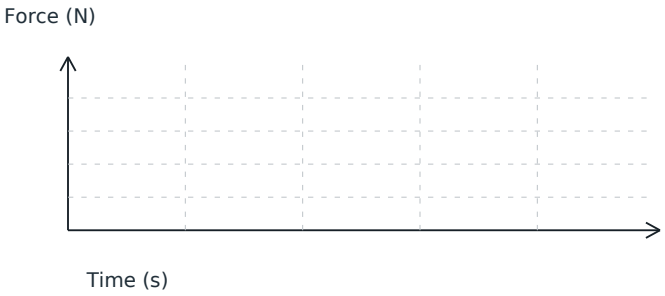
05 Explain your reasoning

List three forms or destinations of energy after a deforming collision. Does “less kinetic energy” mean “energy destroyed”?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A positive triangular force pulse lasts 0.4 s and peaks at 60 N. Find impulse and velocity change for a 0.8 kg object.
Show the signed impulse on the force-time model.



Marzano check - circle: 1 2 3

07 Exit check

At fixed momentum, mass triples. Derive or explain the kinetic-energy multiplier using $K=p^2/(2\,m)$. Use the energy model to show the quantities you compare; label units and any unknowns.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two carts have equal momentum magnitude p. Derive K in terms of p and m, then compare energies if one mass is four times the other. Use the energy model to show the quantities you compare; label units and any unknowns.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3